

Phoenix public parks serve as tree refuges in deprived areas

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1 Introduction

Urban trees, whether in private yards, planted along streets, or in public parks, are a critical component of sustainable cities. The urban forest provides numerous ecosystem services that benefit city residents, including shade, stormwater removal, and air pollution mitigation (CITE). Tree canopy cover (TCC), defined as the area of ground covered by tree leaves, stems, and branches, is especially crucial in hot arid cities like Phoenix, Arizona, where shaded ground can be up to XX C cooler than bare ground (CITE). Adequate shade can be a lifesaving amenity in Phoenix, which experienced XXX heat-related deaths in 2025 (CITE).

Like many urban amenities, tree canopy cover is often not equitably distributed across cities. Several studies have shown that residents of lower socioeconomic status tend to live in “tree deserts,” or neighborhoods with lower TCC (CITE). Some studies also report that neighborhoods with larger populations of ethnic and racial minorities are more likely to be tree deserts (CITE). Wang, et al. (2024) demonstrated that these patterns hold true in Phoenix as well, finding a positive association between TCC and median household income, a negative correlation between TCC and percent Black population, and a negative correlation between TCC and percent Hispanic population at the block group level [1].

As a source of publicly available tree canopy cover, municipal parks could potentially act as “tree oases” and compensate for the lack of tree cover in disadvantaged neighborhoods. However, numerous studies have shown that public parks also tend to be inequitably distributed across cities (CITE). Public parks in XXX and XXX are more concentrated in more affluent neighborhoods with larger White populations, but the evidence for park inequities in Phoenix is mixed. Lara-Valencia (2018) found that Hispanic residents of Phoenix tend to have greater access to municipal parks due to a higher concentration of parks in the urban core [2]. However, Ibes (2015) analyzed access to public parks in Phoenix by type of park and found that “green neighborhood parks” are more concentrated in neighborhoods with higher White populations while “urban core parks” are more often found in areas with lower incomes and more minority residents [3]. Corley (2024) found no association between park acreage and minority status or poverty level at the block group level in Phoenix [4]. The previously mentioned studies on Phoenix park access did not account for spatial autocorrelation of variables, i.e., observations that are spatially proximate tend to be more similar than expected under random distribution (CITE). When Kim, et al. (2023) used spatial regressive models to investigate park access in Phoenix and nearby Mesa, AZ, they found that more affluent neighborhoods had access to greater park acreage within walking distance, but there were no significant correlations between race/ethnicity of residents and the number of parks, size of parks, or distance to the nearest park [5].

All of the aforementioned studies on Phoenix-area parks have used a simple Euclidean distance metric to determine “access,” but this method does not account for the actual distance that residents must travel to visit the parks. The travel distance from a home to a public amenity can vary greatly from the Euclidean distance depending on the layout of the road network (CITE). Most studies consider “park access” to be a binary variable, where a resident has complete access to any parks within a given catchment area and no access to parks outside of the catchment area. However, this approach does not capture the reality of park visitation—people are more likely to visit nearby parks, but will still travel a greater distance to visit larger parks. Kim et al. (2023) found that Phoenix residents tend to favor local parks within 1 km of their neighborhood, but the likelihood that they will travel beyond 1 km to visit larger amenity-rich parks increases as the size of the local parks decreases [6]. Traditional park access research also tends to only address the “supply” side of access by

quantifying park number, area, or amenities (CITE) and ignoring the “demand” side, or the number of people competing for access to the parks. Other studies incorporate the population factor by simply dividing the total park area by the population within a given catchment [7] (CITE MORE?). The Gaussian two-step floating catchment area (G2SFCA) method addresses these shortcomings by incorporating a park supply factor (e.g., total area or amenities), a demand factor (the population of residents with access to the parks), and a travel distance metric that weights closer parks more heavily than distant parks [8].

An accurate assessment of park access using the G2SFCA method depends on the choice of supply and demand origin points used to determine travel distance from residents to parks. Census block group centroids are often used as demand origin points because they represent the “average” location of residents within a reasonably small area (CITE). However, population-weighted block group centroids are less likely to underestimate accessibility [9]. For example, the geographic centroid of a large block group with housing concentrated in one small corner is likely to be located far from areas where most people live, leading to a suboptimal estimate of the travel distance to a public park. Likewise, the geographic centroid of a park is commonly used as the supply origin point, but this choice can provide an inaccurate picture of park access, especially in large parks or long linear parks where the centroid is far from the entrances. Residents also may not be able to access parks at any point along the boundary but are instead limited to a small number of entrances. Using park entrances rather than centroids as the supply origin points in a G2SFCA analysis overcomes these limitations [8].

Residential tree canopy cover provided by trees in neighborhoods and commercial areas likely has a more direct impact on residents’ lives than the tree canopy in public parks since residential trees shade people’s homes, workplaces, and travel routes. But parks may act as tree canopy oases, allowing people to temporarily enjoy the benefits of trees that may be lacking in their daily lives. No studies have investigated the relationship between residential tree canopy cover and access to park tree canopy cover, but a 2013 study by Zhou and Kim investigated neighborhood tree canopy disparities and general walking-distance access to parks in six Illinois cities [10]. They found that in four of the cities, a higher proportion of Black residents was associated with lower tree canopy cover, but there were no disparities in access to parks. However, this study did not quantify the tree canopy cover within parks. This is a serious limitation because parks can vary widely in size and they may not all contain adequate tree cover to provide shade for nearby residents.

Past research on tree canopy equity has identified tree deserts in Phoenix neighborhoods, but no studies to date have isolated residential tree canopy cover from public park canopy cover. There is also a lack of research addressing whether Phoenix public parks act as tree canopy refuges that partially compensate for lack of tree cover in the surrounding neighborhoods. This study attempts to address these research gaps by answering the following questions:

1. To what extent do racial/ethnic composition and socioeconomic deprivation predict residential (non-park) tree canopy cover across Phoenix neighborhoods?
2. To what extent do racial/ethnic composition and socioeconomic deprivation predict public park tree canopy accessibility in Phoenix, and does this relationship diverge from the patterns observed for residential tree canopy?
3. Do Phoenix public parks function as tree canopy refuges — i.e., do neighborhoods with low residential tree canopy have disproportionately high access to park tree canopy — and does this compensatory pattern

hold uniformly across racial/ethnic groups and deprivation levels?

2 Materials and methods

2.1 Study area

The study area is the city of Phoenix in Maricopa County, Arizona, USA. AREA... It has a population of 1.67 million as of 2024, making it the fifth largest city in the country [11]. The Phoenix metro area consists of 26 municipalities with a total of 4.7 million inhabitants. The population of Phoenix is ethnically and racially diverse. Of the block groups included in this study, Hispanic residents make up 39.6% of the population, and Black residents make up 6.8%, according to data from the 2021-2024 5-year American Community Survey (ACS). The median household income is \$9.082259⁴. FIX THIS

Phoenix is located in the Sonoran Desert and has a semi-arid climate with hot summers and mild winters. The average daily mean temperature...

2.2 Block groups and parks

Census block groups were used as the unit of analysis in this study because this is the smallest geographic units for which detailed socioeconomic data are available. A block group is an area defined by the U.S. Census Bureau that encompasses several contiguous city blocks and usually contains between 600 and 3000 residents. Block group boundaries as of 2024 were downloaded from the United States Census Bureau using the tidycensus R package [12].

Phoenix public park boundaries were downloaded from OpenStreetMap [13]. Desert preserves, parks established after 2021, and those missing land cover data were excluded, resulting 177 public urban parks in the final dataset. Private planned community parks were excluded from the analysis as they are often only accessible to neighborhood residents [14].

The City of Phoenix Parks and Recreation Department operates 187 urban parks and desert preserves [15], classified into five categories based on their size and amenities. Each park category has a service area radius that represents the recommended distance from a Phoenix residence to the nearest park of a given type [16]. Park category characteristics are summarized in [Table 1](#).

Table 1: Typical amenities, sizes, and service area radius of Phoenix public park types

Park type	Typical amenities	Size (ha)	Service area (km)
Pocket	Playground; small open area; picnic areas	<0.8	0.8
Linear	Linear open space; walking/biking path; fitness equipment; picnic areas	Various	0.8
Neighborhood	Open space; playground; sports fields; fitness equipment; picnic areas	0.8-11.7	1.6

Community	Amenities found in neighborhood parks, plus: swimming pool; recreational center; dog park; skate plaza; pond	12-20	4.8
Regional	Amenities found in community parks, plus: community/ senior center; sports complex and concessions; skate park	>20	8.0

Source: Article Notebook

2.3 Tree canopy cover

Tree canopy data were extracted from a 2021 land cover raster file created by the University of Vermont Spatial Analysis Lab and provided by American Forests [17]. The imagery has a spatial resolution of 0.3 m and includes most of Phoenix except for large undeveloped desert preserves and some urban land on the periphery of the city. A total of 1017 block groups had tree canopy cover data available. The percentage of total tree canopy cover (TTCC) per block group was calculated using the zonal histogram tool in QGIS. TTCC represents the total amount of tree canopy cover within a block group, including street trees and those in private yards, businesses, and parks. Public park polygons were then clipped from the land cover raster to calculate the percentage of tree cover on the non-park land within block groups, known as the residential tree canopy cover (RTCC). [Figure 1](#) shows the effect of trees within a public park on the tree canopy cover within a block group. Although RTCC includes trees within non-residential areas such as schools, churches, and parking lots, they are considered “residential” in this study since they are not located in public parks, and they have a more direct impact on residents’ daily lives.

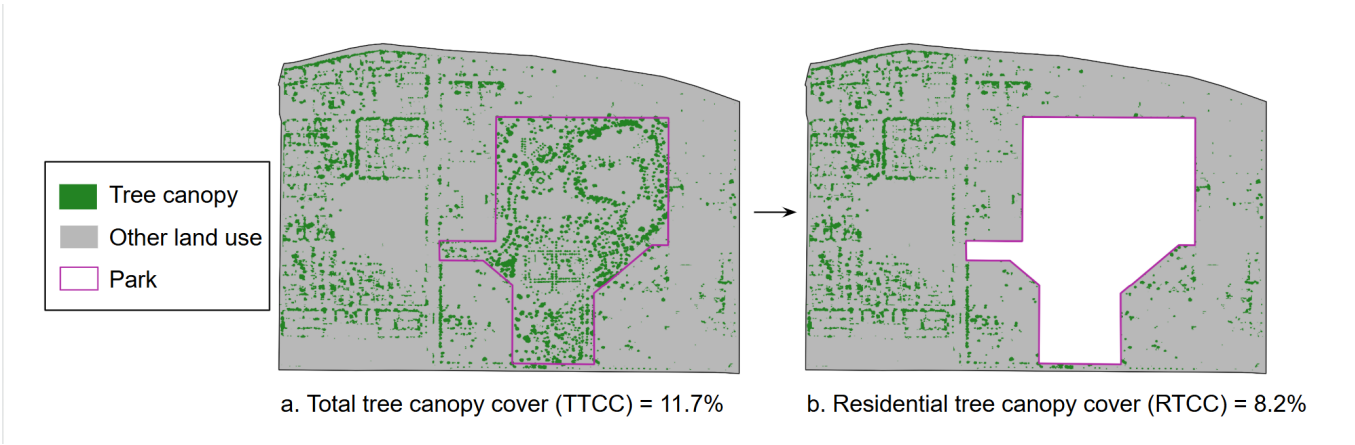


Figure 1: The effect of public park tree canopy cover on total tree canopy cover

The Mann-Whitney *U* test was performed to determine whether TTCC within block groups is affected by the presence of public parks. The median TTCC value of the 176 block groups with parks was compared to the TTCC value of the 841block groups without parks, and the *U* statistic was calculated.

The Wilcoxon signed-rank test was performed on the block groups that contain public parks to determine whether the parks contribute to the total tree canopy cover within those block groups.

2.4 Demographic factors

Total population and race/ethnicity data at the block group level were sourced from the 2024 American Community Survey (ACS) 5-year estimate [18] using the tidycensus R package. These factors were used to calculate the percentage of Black and Hispanic residents per block group. Population estimates from the 2024 ACS 5-year estimate were also downloaded for individual Census blocks to be used for calculating population-weighted block group centroids.

The area deprivation index (ADI) at the block group level was acquired from the University of Wisconsin School of Medicine Public Health [19]. The ADI is a validated measure of socioeconomic status based on 17 indicators of income, education, employment, and housing quality [20]. The raw ADI score of a block group is compared against all other block groups in the nation and ranked on a percentile scale from 1-100, where 1 indicates the lowest level of disadvantage and 100 indicates the highest disadvantage. Block groups containing predominantly group housing (college dormitories or jails) do not have ADI data available, so they were excluded from the analysis. The resulting dataset contained 988 block groups used in the demographic analysis. [Figure 2](#) shows the locations of block groups with land cover and/or demographic data available and the types of public parks within the block groups.

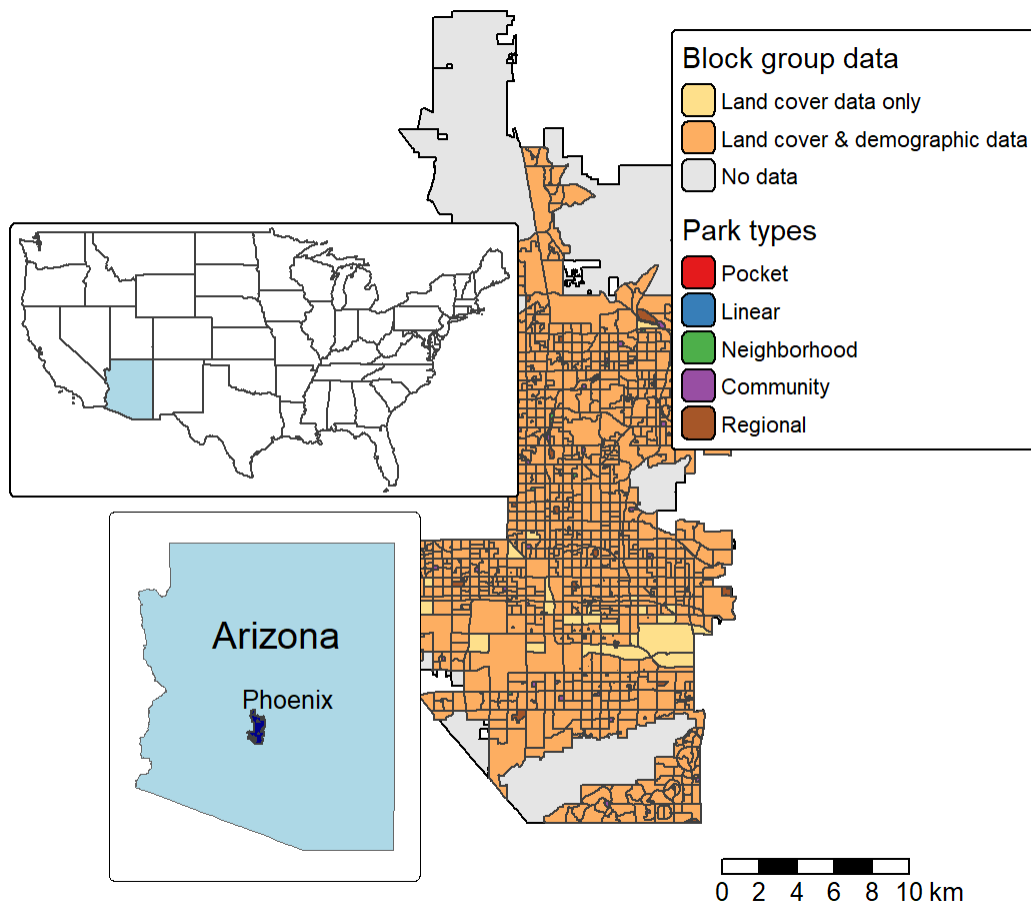


Figure 2: Block groups and public parks in Phoenix, Arizona

Median household income is often used as a predictor variable in green space access research [7], [10], but it was excluded from this study due to its high correlation with ADI rank ($r = -0.70$) and the presence of 50 block groups with suppressed income estimates in the ACS data.

2.5 Spatial access to park tree canopy cover

The Gaussian two-step floating catchment area (G2SFCA) method was used to determine Phoenix residents' spatial access to tree canopy cover in public parks. This method accounts for both the supply of public amenities and demand on those amenities when calculating accessibility. In previous studies of green space equity, park size and amenities have often been used as the supply term (CITE). The population of areas surrounding an amenity represents the demand on the amenity. More people living near a park leads to greater competition for the park's amenities, including tree canopy cover.

Park tree canopy area in hectares (PTCA) was used as the supply term in this study and was calculated by multiplying the percent tree canopy cover of a park by its total area ([Table 2](#)). Since PTCA values are affected by both park size and percent tree canopy cover, residents living near several small tree-dense parks may have access to more tree canopy than residents near a single large park with scant tree cover.

Table 2: Total park area and park tree canopy cover area in Phoenix public parks

	Mean	SD	Median	Min	Max
Total park area (ha)	8.30	12.51	4.46	0.08	115.25
Park tree canopy area (ha)	1.08	1.53	0.60	0.00	14.13
<i>N</i> = 177					

The original 2SFCA, developed by Luo and Wang (2003) [21], considers an amenity to be fully accessible if it lies within a given travel time or distance from a residential area and fully inaccessible if it lies outside this catchment area. However, the decision to visit an amenity is not dichotomous; for example, a resident may choose to travel farther to visit a park with more tree canopy even though there are other parks nearby that provide less shade. The Gaussian 2SFCA method used in this study improves on the original method by applying a continuous weighting function to the travel distance, where nearby parks are given more weight than those farther away within the same catchment area. Residents who choose more distant parks will thus encounter greater travel friction [22].

The first step in the G2SFCA method is to compute the ratio of park tree canopy area to the surrounding block group population at each park location. The population-weighted centroid of block groups was used as the population location (k) in calculating the distance between block groups and parks. This point represents the mean location of residents within the block group. Populations at each block group location k are weighted using a Gaussian function. This ratio is calculated as:

$$R_j = \frac{S_j}{\sum_{k \in d_{kj} \leq d_0} G(d_{kj}, d_0) P_k}, \quad (1)$$

where P_k is the population at block group k whose population-weighted centroid lies within the catchment (i.e., $d_{kj} \leq d_0$) of a park at location j ; d_{kj} is the travel distance between block group k and park j ; S_j is the supply capability of park j , i.e., the tree canopy area in hectares; G is the friction-of-distance, calculated below:

$$G(d_{kj}, d_0) = \begin{cases} \frac{e^{-(1/2) \times (d_{kj}/d_0)^2} - e^{-(1/2)}}{1 - e^{-(1/2)}}, & \text{if } d_{kj} \leq d_0 \\ 0, & \text{if } d_{kj} > d_0 \end{cases} \quad (2)$$

The second step in G2SFCA is to search all parks l within the threshold distance d_0 from block group i , thus creating the catchment for the population at i . Each R is discounted using the Gaussian function G . The discounted R values within the catchment area of i are summed to obtain the spatial accessibility of park tree canopy area at block group i as follows:

$$A_i = \sum_{l \in d_{il} \leq d_0} G(d_{il}, d_0) R_l, \quad (3)$$

where l denotes all parks within the catchment of block group i , and all other notations are the same as in [Equation 1](#). The park tree canopy accessibility score (A_i) represents the amount of park tree canopy cover (in hectares) available to each resident in a given block group. Raw accessibility scores are very small values and thus were re-scaled by a factor of 1,000,000 for reporting purposes; the re-scaled values are hereafter referred to as park tree access scores (PTAS).

Public park access studies often use the park centroid as the destination point. However, this can bias the results against large parks since users may have to walk long distances from a park entrance to the centroid. Parks may also have multiple points of entry that are separated by significant distances [23]. This study used park entrances, determined by visual examination using QGIS, as destination locations for parks larger than 2 ha. For smaller parks, the centroid was used since travel time would be minimally affected by the access point location.

Travel distance (d_{kj}) from block group centroids (k) to park entrances (j) was computed using the driving profile in the osrm R package [24]. This method uses the Open Source Routing Machine to identify the shortest driving path between any two points based on the OpenStreetMap road network. A distance matrix of all block group population-weighted centroids to all park entrances was created and then limited to the distance from a block group to only the nearest entrance within any given park.

Park catchment size (d_0) was determined by the service area radius assigned to each park category by the City of Phoenix [25]. Pocket and linear parks were assigned a threshold of 0.8 km (0.5 mi), neighborhood parks 1.6 km (1 mi), community parks 4.8 km (3 mi), and regional parks 8.0 km (5 mi). This aligns the park catchment values with the park accessibility goals of the City of Phoenix.

The G2SFCA method is vulnerable to the “edge effect” problem wherein phenomena contributing to the spatial patterns of an analysis unit may lie outside of that unit [26]. In other words, accessibility scores for block groups on the periphery of Phoenix may be negatively impacted if block groups from surrounding cities are not included in the analysis. This is because residents of these other cities can use Phoenix parks, thus diluting the amount of tree cover available to Phoenix residents. The edge effect was accounted for in the demand portion of the method because the populations of 682 block groups from surrounding cities that lie within a park catchment area were included in the calculation of the supply/demand ratios (R_j and R_l). The edge effect can also impact the supply of park tree canopy area since Phoenix residents can access parks in other cities. However, this is not relevant to the current study because the purpose was to measure the access of residents

only to those parks maintained by the City Phoenix. Hence, public parks located in nearby cities were not included in the analysis.

2.6 Regression analysis of tree canopy cover disparities

Evaluating demographic disparities in block group-level tree canopy cover began with ordinary least squares (OLS) regression models using RTCC and PTAS as the predictor variables and percent Hispanic population, percent Black population, and ADI national percentile rank as the independent variables. An initial OLS model using the untransformed PTAS score showed non-normal residuals (Shapiro-Wilk $W=0.843$, $p<.001$) with visible right skew in a Q-Q plot. A square root transformation was applied to PTAS and the model re-estimated; residual normality improved substantially ($W=0.955$) but remained significantly non-normal ($p<.001$). Percent White population was excluded from the analyses due to collinearity with the percent Hispanic population. Percent of the population under 18 years old and over 64 years old were considered as predictors, but both had an insignificant impact on the OLS results.

Demographic and environmental variables often exhibit spatial autocorrelation, meaning the residuals in a given block group are likely to be similar to their surrounding block groups. The presence of spatial autocorrelation violates the assumption of independent residuals necessary for OLS regression, necessitating the use of spatially aware regression models. The global Moran's I index was calculated to measure the overall spatial autocorrelation of the residuals of both OLS regression models using the following formula:

$$I = \frac{N}{S_0} \frac{\sum_{i=1}^N \sum_{j=1}^N w_{ij}(x_i - \bar{x})(x_j - \bar{x})}{\sum_{i=1}^N (x_i - \bar{x})^2}, \quad (4)$$

where N is the number of block groups indexed by i and j , x is the variable of interest (i.e., OLS residuals), $\bar{x}$ is the mean of x , w_{ij} is a spatial weights matrix that represents the degree of spatial proximity between block groups i and j . The distance matrix was calculated based on the eight nearest neighbors of each block group. This method allowed for all block groups to be included in the computation, resulting in no isolated “islands” of block groups. A Monte Carlo simulation with 999 iterations was used, and the p -value was set at 0.01 to evaluate the significance of the global Moran's I index. The value of Moran's I ranges from -1 to 1; positive values indicate clustering, negative values indicate dispersion, and 0 indicates a random distribution of residuals.

Since spatial autocorrelation was present in the residuals of the OLS models for both dependent variables, spatial regression models were warranted. Lagrange Multiplier (LM) tests were used to determine the appropriate spatial regression specification for each outcome, following the decision rule recommended by Anselin and Serge [27]: if the robust LM error statistic is significant and the robust LM lag statistic is not, a spatial error model is selected; if robust LM lag is significant and robust LM error is not, a spatial lag model is selected. LM tests indicated that a spatial error specification was appropriate for both RTCC and PTAS (see [Section 3](#)).

In a spatial error model, spatial dependence is captured in the error term rather than the outcome itself. This specification is appropriate when neighboring block groups have similar residuals due to unmeasured place-based factors that cluster spatially and affect tree canopy cover, such as housing age or homeowners association planting guidelines. By modeling this spatial structure as an error term, the spatial error model isolates the independent effects of racial/ethnic composition and socioeconomic disadvantage on tree canopy

outcomes. The spatial error model takes the following form:

$$y = X\beta + u, \quad (5)$$

where y is the outcome (RTCC or PTAS), X represents the predictor variables (percent Hispanic population, percent Black population, and ADI rank), β represents the coefficients, and u is an error term modeled as:

$$u = \lambda Wu + \epsilon, \quad (6)$$

where λ is the spatial error coefficient, Wu is the weighted average of neighboring errors based on a spatial weights matrix of the eight nearest neighbors, and ϵ is a random independent error term.

2.7 Compensatory tree canopy access (need better title)

After separately identifying spatial patterns in residential tree canopy cover and public park tree access scores, block groups were classified as either high RTCC or low RTCC. Those with a RTCC of 15% or higher were classified into the high RTCC category and the rest were considered low RTCC. This threshold is based on the recommendation of the non-profit organization American Forests that desert cities should maintain a 15% tree canopy cover to achieve tree equity [28]. In the absence of a clear policy guideline for park tree access score, block groups with a PTAS below the median were categorized as low PTAS, and those at or above the median were categorized as high PTAS. The two factors were then combined to create four tree canopy access categories: a) high RTCC/high PTAC (HH), b) high RTCC/low PTAS (HL), c) low RTCC/high PTAS (LH), or d) low RTCC/low PTAS (LL).

Multinomial logistic regression was used to examine whether the percent Hispanic, percent Black, or ADI rank predicted a block group's membership in each tree canopy access category relative to the high RTCC/high PTAS condition (HH). Given that spatial autocorrelation was detected in the residuals of both the RTCC and PTAS regression residuals, it was expected that the residuals of the tree access category logistic regression model might exhibit similar spatial dependence. However, global Moran's I cannot be calculated directly on the residuals of a multinomial logistic model, since the outcome comprises four unordered categories rather than a single continuous or binary residual vector. To address this, each tree canopy access category (HH, HL, LH, LL) was recast as a binary outcome (one-vs-rest) and modeled separately via logistic regression on the same three demographic predictors. Global Moran's I was then calculated on the residuals of each binary model to test for spatial autocorrelation.

Significant spatial autocorrelation was detected on the residuals of all four binary regression models (see [Section 3](#)), so the Moran eigenvector spatial filtering approach was used to incorporate spatial predictors into the models [29]. This approach decomposes a spatial weights matrix into orthogonal eigenvectors, each representing a distinct spatial pattern at a different scale. These eigenvectors are incorporated into regression model as additional predictors, thereby absorbing the spatial signal that would otherwise be part of the model's error term. Eigenvector selection was implemented using the spatialreg R package [30], using the eight nearest-neighbors matrix used previously. Each eigenvector was tested iteratively, and those that significantly reduced Moran's I ($\alpha = 0.08$) were retained. The selected eigenvectors were added as additional covariates to each binary logistic model, and global Moran's I was recalculated on the resulting residuals to assess the degree to which spatial autocorrelation had been reduced.

3 Results

3.1 Impact of public parks on tree canopy cover

Percent total tree canopy cover (TTCC) is the amount of tree canopy cover in a block group including the tree cover provided by public parks. Across the entire city, the 176 block groups that contain parks have a median TTCC of 10.95, while the 841 block groups without parks have a median TTCC of 11.17. This difference is non-significant (Mann-Whitney $U = 78897$, $p = 0.168$), suggesting that parks are evenly distributed across the canopy spectrum rather than concentrated in already-green neighborhoods.

Percent residential tree canopy cover (RTCC) is the tree cover in a block group excluding the tree cover provided by public parks. TTCC and RTCC were compared in the 176 block groups that contain parks to determine whether the parks significantly contribute to block group-level tree canopy cover in Phoenix. The median TTCC was 10.20%, while RTCC was higher at 10.95%. This difference was statistically significant (Wilcoxon signed-rank test, $V = 11271$, $p < 0.001$), confirming that parks meaningfully contribute to tree canopy cover when they are present in a neighborhood. [Table 3](#) (mention how parks sometimes reduce canopy cover)

Table 3: Total tree canopy cover and residential tree canopy cover in block groups that contain parks

	Mean	SD	Median	Min	Max
Total tree canopy cover (%)	11.47	4.19	10.95	3.65	26.28
Residential tree canopy cover (%)	11.25	4.36	10.20	3.49	27.50
Difference between TTCC and RTCC	0.23	0.74	0.12	-2.59	3.94
<i>N</i> = 176 block groups					

3.2 Regression analysis

[Table 4](#) presents descriptive statistics for the block group characteristics and outcome variables (RTCC and PTAS) used in the regression analyses. Block group population is included for reference, but it was not used as a predictor variable. Maps of RTCC and PTAS are shown in [Figure 3](#), and maps of the three predictor variables (percent Hispanic population, percent Black population, and ADI rank) are depicted in [Figure 4](#).

Table 4: Descriptive statistics of the dataset

	Mean	SD	Median	Min	Max
<i>Block group characteristics</i>					
Total population	1,597	710	1,499	102	6,221
Hispanic population (%)	39.7	28.7	33.9	0.0	99.3

Black population (%)	6.8	9.5	3.0	0.0	59.1
ADI rank	40	19	40	1	100

Outcome variables

Residential tree canopy cover (%)	11.9	4.3	11.1	2.6	28.2
Park tree canopy access score	105.6775	58.48181	97.3453	1.445721	567.9618

N = 988 block groups

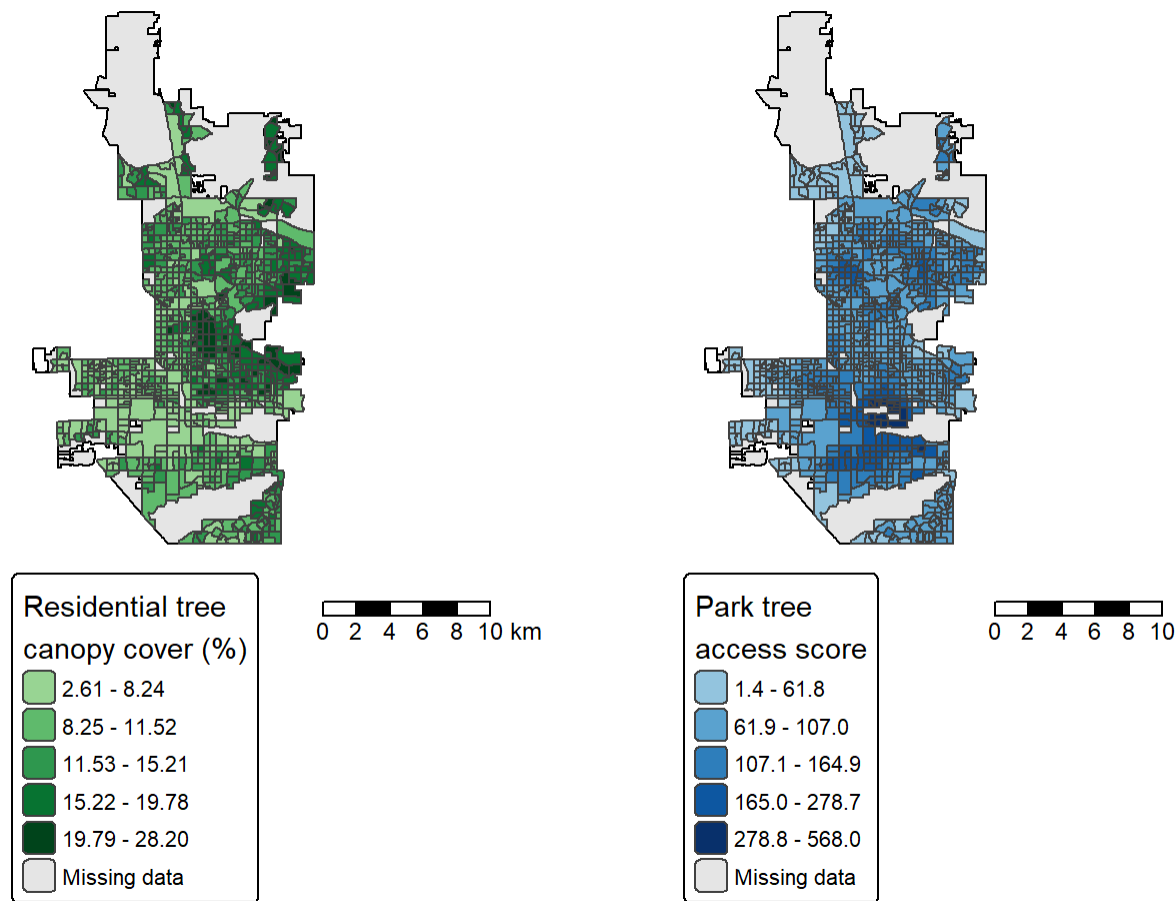
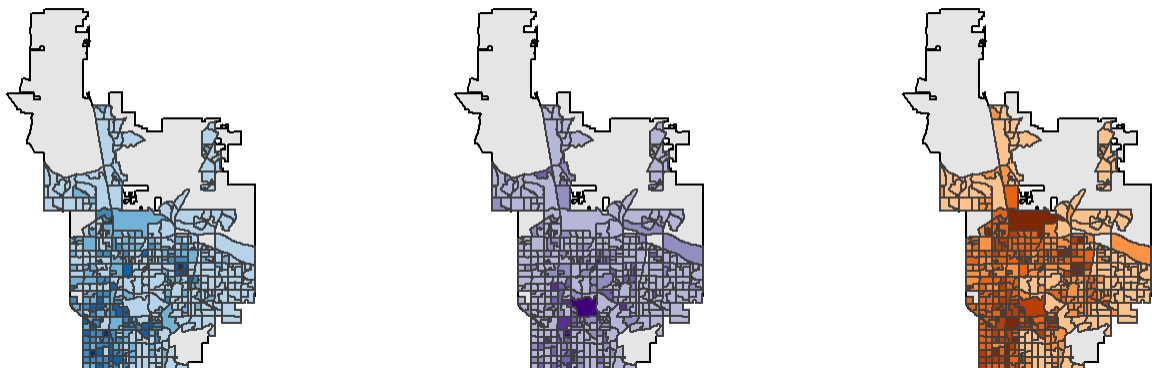


Figure 3: Residential tree canopy cover (%) and park tree access score of Phoenix block groups



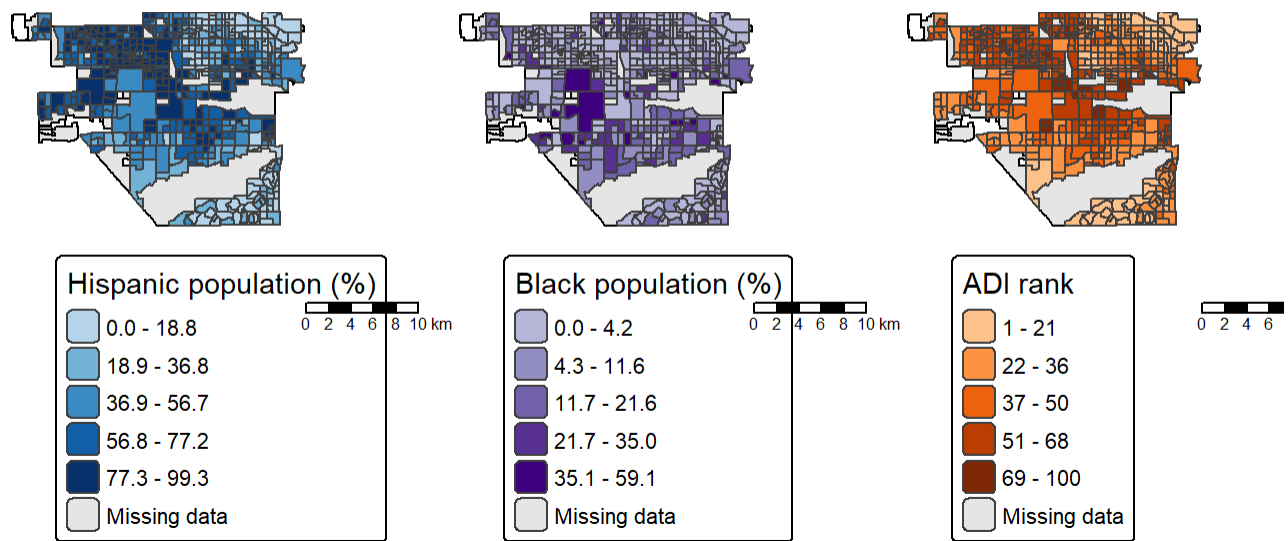


Figure 4: Maps of Hispanic population (%), Black population (%), and ADI national percentile rank in Phoenix block groups

Based on the OLS model, percent Hispanic population, percent Black population, and ADI rank were all negatively associated with RTCC ($p < 0.001$; see [Table 5](#)). For every 1% increase in the Hispanic population, residential tree canopy cover within a block group is expected to decrease by 3.5%; a 1% increase in the population of Black residents is associated with a 7.3% decrease in RTCC; and a 1-point increase in the ADI rank is associated with an 8.5% decrease in RTCC.

The percentage of Hispanic residents in a block group is not significantly associated with the square root of PTAS, but there was a significant positive association between the square root of PTAS and both Black resident percentage ($p < 0.05$) and ADI rank ($p < 0.001$; see [Table 5](#)).

These findings suggest that while neighborhoods with a higher Black population and lower socioeconomic status have lower tree canopy cover provided by trees on private property, street trees, and trees in private parks, the public parks near these neighborhoods provide greater access to tree canopy cover than those near neighborhoods with lower Black populations and higher socioeconomic status.

Global Moran's I was significant ($p < 0.001$) for both the RTCC and PTAS residuals from the OLS models. The RTCC index was 0.221, indicating moderate spatial clustering of residential tree canopy values, and the PTAS index was 0.418, indicating strong spatial clustering of park tree access scores.

Spatial error model results are presented in [Table 5](#) along with OLS estimates for comparison. Similar to the OLS model, all three predictor variables are negatively correlated with RTCC, indicating that block groups with larger Hispanic populations, larger Black populations, or higher ADI rank are more likely to have lower residential tree canopy cover.

The spatial error model for the square-root transformed PTAS also indicates that there is no correlation between park tree access and Hispanic population percent. The coefficient of 0.019 for the Black population percent (significant only at the $p < 0.1$ level) suggests that there may be a slight increase in park tree access as the Black population increases. The significant positive relationship between PTAS and ADI rank indicates that the park tree access score increases as socioeconomic status decreases.

The lambda (λ) term indicates how neighboring observations influence one another's errors. The positive λ in

both spatial error models indicates that unmeasured error in one block group significantly impacts the errors in adjoining block groups for both RTCC and PTAS.

Table 5: Results of OLS and spatial error regression models for RTCC and PTAS

	RTCC models		√PTAS models	
	OLS	Spatial error	OLS	Spatial error
(Intercept)	20.703***	17.199***	5.227***	8.541***
Hispanic population (%)	-0.037***	-0.040***	-0.007†	0.000
Black population (%)	-0.064***	-0.069***	0.011	0.019*
adi	-0.065***		0.046***	
ADI rank		-0.080***		0.027***
lambda		0.462***		0.731***
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$				
$N = 988$				

3.3 Demographic inequities in tree canopy access

Block groups were organized into one of four tree canopy access categories based on their RTCC and PTAS values (Table 6). The spatial distribution of each category is illustrated in Figure 5. The least common category is HH, or neighborhoods with both abundant residential tree canopy and higher than average access to park tree canopy. These block groups are mostly confined to isolated pockets in the northeastern part of Phoenix. Block groups in the HL category have lower than average access to park trees, but the residential tree canopy is abundant. The most common category is LH, representing block groups with low residential tree canopy but high access to park canopy cover. These are low-canopy neighborhoods where public parks provide a tree canopy refuge to the residents. They are located mostly in the center and northern portions of the city where most of the public parks are situated. The remaining block groups belong to the LL category, or “tree canopy deserts,” lacking both residential tree canopy cover and access to public park tree canopy. They tend to be found on the periphery of the city.

Table 6: Tree canopy access categories

Category	Rule	Description	n Block groups (%)
HH	High RTCC + High	Abundant residential and park tree	29 (9.0)

Category	PTAS	canopy	Count (Percentage)
HL	High RTCC + Low PTAS	Self-sufficient residential tree canopy	122 (12.3)
LH	Low RTCC + High PTAS	Park tree canopy refuge	404 (40.9)
LL	Low RTCC + Low PTAS	Tree canopy desert	373 (37.8)

N = 988 total block groups

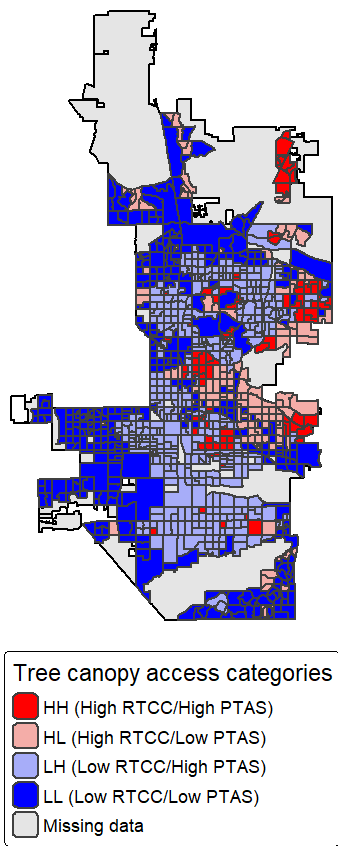


Figure 5: Tree canopy access categories of Phoenix block groups

The multinomial logistic regression model estimates the odds that a particular demographic factor (Hispanic population, Black population, or ADI rank) predicts whether a block group belongs to either the HL, LH, or LL tree access category rather than the baseline HH category. The results show that block groups are equally likely to be in the HL category compared to the HH category, regardless of the demographic makeup of the block group. In other words, neighborhoods with high residential tree canopy cover and high park tree canopy access are not demographically distinct from those with high residential canopy and low park tree access ([Table 7](#)).

Table 7: Multinomial logistic regression odds ratios for canopy access categories

Predictor	HL		LH		LL	
	OR [†]	95% CI	OR	95% CI	OR	95% CI
Hispanic population (%)	1.00	0.98, 1.02	1.02*	1.00, 1.03	1.01	1.00, 1.03
Black population (%)	0.99	0.94, 1.05	1.05*	1.01, 1.10	1.04 [†]	1.00, 1.08
ADI rank	0.99	0.96, 1.01	1.08**	1.05, 1.10	1.06**	1.03, 1.08

Reference category: HH (High RTCC/High PTAS)

[†] $p < 0.1$, * $p < 0.05$, ** $p < 0.01$

$N = 988$: HH = 89 (9.0%), HL = 122 (12.3%), LH = 404 (40.9%), LL = 373 (37.8%)

[†] OR: Odds ratio; CI: Confidence interval

All three demographic factors were significantly associated with membership in the LH category ($p < 0.05$). For every 1% increase in the Hispanic population, there is a 2% increase in the odds that a block group belongs to the LH category vs. the HH category. A 1% increase in Black population is associated with a 5% increase in the odds of belonging to the LH category. A one-point increase in ADI national percentile rank is associated with an 8% increase in the odds of being in the LH category.

For the LL tree access category, only ADI rank is significant ($p < 0.001$). For every one-point increase in ADI rank, there is a 6% increase in the odds that a block group belongs to this category compared to the HH category.

Higher ADI rank was associated with greater odds of belonging to both low RTCC categories, LH and LL, relative to HH. This supports the previous finding that low socioeconomic status is linked to reduced residential tree canopy cover in Phoenix. But there is no clear relationship between the socioeconomic level of a block group and its likelihood of having greater access to public park tree canopy cover.

The higher odds of a block group with a larger Black or Hispanic population belonging to the LH category suggests that these neighborhoods are more likely to have parks that act as tree canopy refuges, providing higher park tree canopy access despite having low residential tree canopy cover.

Because spatial autocorrelation was detected in the one-vs-rest binary logistic models underlying the four-category canopy access typology (Moran's I : HH = 0.140, HL = 0.144, LH = 0.267, LL = 0.232; all $p < 0.001$), Moran eigenvector spatial filtering was applied to the models. Selected eigenvectors (HH: 9, HL: 14, LH: 30, LL: 23) were added as covariates to each binary model, substantially reducing spatial autocorrelation across all four categories. Moran's I was reduced by 76-85% in each model (HH = 0.033, HL = 0.030, LH = 0.040, LL = 0.034), though modest statistically significant spatial clustering remained in each case ($p < 0.05$). This result is consistent with spatial structure operating at a scale broader than the local nearest-neighbor weighting could capture.

[Table 8](#) reports the β coefficients for each tree canopy access binary logistic models before and after spatial filtering. Both the HH and HL categories were negatively associated with ADI national percentile rank, while the LH and LL categories were positively associated with ADI rank. These associations were significant both before and after spatial filtering, indicating that the relationship between ADI and canopy access category is robust to the presence of spatial autocorrelation in the model residuals. Significance levels for most other predictor β terms also remained stable after filtering, with two exceptions: for the HH category, the negative associations with Hispanic population and Black population moved from marginal significance ($p < 0.1$) to significance at $p < 0.05$; and for HL, the negative association with Hispanic population shifted from non-significant to significant at $p < 0.05$, while the association with Black population strengthened from $p < 0.05$ to $p < 0.01$.

Table 8: Binary logistic regression results for canopy access categories before and after spatial filtering

Term	β (unfiltered)	β (filtered)
<i>HH</i>		
Intercept	-0.303	-0.778**
Hispanic population (%)	-0.013 [†]	-0.018*
Black population (%)	-0.035 [†]	-0.041*
ADI rank	-0.044***	-0.039***
<i>HL</i>		
Intercept	0.577**	0.628*
Hispanic population (%)	-0.01	-0.021**
Black population (%)	-0.043*	-0.066**
ADI rank	-0.065***	-0.068***
<i>LH</i>		
Intercept	-2.13***	-2.533***
Hispanic population (%)	0.007*	0.006
Black population (%)	0.019**	0.025**
ADI rank	0.032***	0.038***
<i>LL</i>		
Intercept	-0.76***	-1.151***

Hispanic population (%)	0.001	0.005
Black population (%)	0.005	0.001
ADI rank	0.004	0.008

† $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

4 Discussion

This study examined the ability of Phoenix residents to access both residential tree canopy cover and public park tree canopy cover.

4.1 Public park contribution to tree canopy cover

Public parks contribute to the total tree canopy cover found within Phoenix neighborhoods. Although the average increase in tree cover is small, it is statistically significant. On average, Phoenix neighborhoods that contain parks have higher tree canopy cover than they would have if the parks were not present.

4.2 Inequitable distribution of tree canopy cover

This study confirms that tree canopy cover is inequitably distributed across Phoenix. Previous studies have shown that total tree canopy cover is highest in block groups with lower racial and ethnic minorities and higher socioeconomic status. This study adds to these findings that this pattern also applies to residential tree canopy cover, excluding the tree canopy provided by public parks.

This is the first study to analyze access to public park tree canopy cover in Phoenix. It is also the first study to use the Gaussian two-step floating catchment area (G2SFCA) method to analyze access to tree canopy cover in parks. This method is superior to traditional measures of park access that only consider presence/absence of parks and use a binary distance metric to determine access. G2SFCA accounts for both tree canopy area and population-based demand on the tree canopy while also applying distance-based weighting to the calculation of distance from block groups to parks. The use of actual park entrances for determining distance-to-park is also an improvement over centroid- or perimeter-based methods that may over- or under-estimate the true distances from residential areas to the parks. Zhang et al. (2023) found that the use of centroids in a park access study underestimated the disparities in accessibility compared to the use of park entrances [8].

G2SFCA could be applied to other cities to determine whether access to public tree cover is equitably distributed.

Park tree access score is better than amount of tree canopy area available to each block group because it takes population demand into account. Other studies focus on access to parks but don't consider park tree cover as an amenity. Some studies

4.3 Limitations and future research

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